

Booking Cancellation Logic

A complete, human-readable guide to when, how, and why cancellation is allowed or blocked — for both patients and caregivers

This document explains every rule that governs booking cancellation in Nursify. It covers which booking states permit cancellation, which are permanently blocked, what happens differently for a patient versus a caregiver, how refunds and penalty flags work, what error messages the system returns, and what each side of the app should do at every stage.

The goal is for any developer, product manager, or tester to fully understand the cancellation system without needing to read source code.

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1. Why Cancellation Needs Strict Rules

In a healthcare services marketplace like Nursify, a cancellation is not just a button press — it has real consequences for real people. A caregiver may have already arranged transport, left their home, or turned down other work to fulfil a booking. A patient may be in genuine need of care and relying on the caregiver's arrival.

The cancellation rules in Nursify are designed to balance two things fairly: giving patients and caregivers a reasonable window to change plans without penalty, while protecting both parties once a commitment has been physically acted upon.

The core principle:

Cancellation is free and easy before the caregiver starts travelling. Once the caregiver marks themselves as 'on the way', cancellation is permanently blocked for both sides. The service must be completed or resolved through the admin.

2. The Booking Lifecycle — A Quick Recap

Every booking in Nursify moves through a fixed sequence of statuses. Understanding these statuses is essential to understanding the cancellation rules, because each rule is tied directly to the status the booking is currently in.

Status	What it means
pending	The patient has sent a booking request. The caregiver has not yet responded. No money has changed hands.
approved	The caregiver has accepted the request. The patient must now choose and confirm a payment method.
confirmed	The patient has confirmed how they will pay. If paying by card, payment has been processed. The booking is locked in.
on_the_way	The caregiver has started travelling to the patient. Live GPS tracking is now active. This is the point of no return.
arrived	The caregiver has physically reached the patient's location.
service_started	The caregiver has begun delivering the service.
service_completed	The caregiver has finished the service. The patient must now confirm completion.
completed_confirmed	The patient has confirmed completion. The booking is fully closed. Reviews may now be submitted.
rejected	The caregiver declined the pending request before accepting.

cancelled	Either the patient or caregiver cancelled the booking using the rules described in this document.
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3. The Point of No Return

The most important concept in Nursify's cancellation logic is the point of no return. This is the moment when the booking status changes from 'confirmed' to 'on_the_way'.

Once a caregiver marks themselves as 'on the way', cancellation is permanently blocked for both the patient and the caregiver. No exceptions. The system will reject any cancellation request with a clear error message.

Why is this the cutoff point?

- The caregiver has physically committed. They have left their location and are actively travelling. Cancelling at this point causes real inconvenience and potential financial loss.
- GPS tracking has started. The caregiver's app begins broadcasting live location data the moment 'on the way' is set. The system is now in an active service state.
- The patient is expecting arrival. Allowing the patient to cancel now — after a confirmed and paid booking — would undermine trust in the platform.
- The state cannot be cleanly undone. The caregiver has progressed through confirmed into an active travel state, which cannot be reversed without side effects.

The cancellation window at a glance:

<i>pending</i>	<i>approved</i>	<i>confirmed</i>	<i>on_the_way</i>	<i>arrived</i>	<i>started</i>	<i>completed</i>
<i>Free cancel</i>	<i>Free cancel</i>	<i>Cancel (conditions)</i>	<i>BLOCKED</i>	<i>BLOCKED</i>	<i>BLOCKED</i>	<i>BLOCKED</i>

Green = cancellation allowed. Amber = allowed with conditions. Red = permanently blocked.

4. Patient Cancellation — Full Rules

A patient is the person who requested the booking. Here is exactly what they can and cannot do at every stage of the booking lifecycle.

When the booking is in 'pending' status

The patient can cancel freely. The caregiver has not yet responded, no payment has been made, and no commitment exists beyond the initial request. The cancellation is immediate. No reason is required from the patient.

When the booking is in 'approved' status

The caregiver has accepted but the patient has not yet paid. The patient can still cancel freely. The caregiver will be notified the next time their app polls for updates. No reason is required, and no penalty applies.

When the booking is in 'confirmed' status

This is the most sensitive cancellation scenario for a patient. They have confirmed their payment method, and if they paid by card, money has actually been charged. The caregiver is preparing to travel but has not yet left.

A patient CAN cancel a confirmed booking, but this is treated as a significant action. If they paid by Mastercard, a refund must be processed. The system marks the booking with refundStatus 'pending' and triggers the refund through the payment gateway. If they paid by cash, no financial action is needed.

When the booking is in 'on_the_way' status or any later status

The patient CANNOT cancel. The system returns: 'Cancellation is not allowed — the caregiver is already on the way to you.' This hard block applies to all states from on_the_way through to completed_confirmed.

Patient cancellation summary:

Booking status	Can patient cancel?	Conditions
pending	Yes	None — completely free
approved	Yes	None — completely free
confirmed	Yes	Refund triggered if card was used
on_the_way	No	Caregiver en route — hard block
arrived	No	Caregiver on-site — hard block

service_started	No	Service in progress — hard block
service_completed	No	Only confirm completion is available
completed_confirmed	No	Booking already closed

5. Caregiver Cancellation — Full Rules

Caregiver cancellation works differently from patient cancellation in three important ways: caregivers cannot cancel a pending booking (they must use 'reject' instead), they are always required to provide a written reason, and cancelling a confirmed booking raises a penalty flag.

When the booking is in 'pending' status

A caregiver CANNOT cancel a pending booking. Pending bookings have a separate 'reject' action specifically for this purpose. If a caregiver calls the cancel endpoint on a pending booking, the system returns: 'Use the reject action to decline a pending booking request.'

Why is 'reject' different from 'cancel'?

- Reject means: I choose not to accept this request. It is a decision made before any commitment.
- Cancel means: I accepted this request but I am now backing out of that commitment. This carries more weight and is tracked differently.
- Keeping them separate allows admins to distinguish caregivers who often decline requests from those who back out after accepting.

When the booking is in 'approved' status

The caregiver accepted the booking but the patient has not yet paid. The caregiver can cancel, but they must provide a written reason. The system will reject the request if no reason is given. The patient will be notified and returned to the caregiver discovery screen.

When the booking is in 'confirmed' status

This is the most serious cancellation a caregiver can make. The patient has already paid or committed to cash payment, and is expecting the caregiver to arrive. Cancelling now causes direct disruption to the patient.

A caregiver CAN cancel a confirmed booking, but it triggers a PENALTY FLAG. The booking is recorded with penaltyFlag set to true and the caregiver's reason is stored. This flag is visible to administrators. Caregivers who repeatedly cancel confirmed bookings can be reviewed or removed from the platform. A written reason is mandatory — the system will not process the cancellation without one.

When the booking is in 'on_the_way' status or any later status

The caregiver CANNOT cancel. The system returns: 'You cannot cancel while en route. Please complete the visit.' Once a caregiver has marked themselves as on the way, they are committed to completing the service. This protects the patient who is now waiting.

Caregiver cancellation summary:

Booking status	Can caregiver cancel?	Conditions
pending	No — use reject	Reject action is the correct route
approved	Yes	Written reason is mandatory
confirmed	Yes	Reason mandatory and penalty flag raised
on_the_way	No	En route — hard block
arrived	No	On-site — hard block
service_started	No	Service in progress — hard block
service_completed	No	Awaiting patient confirmation only
completed_confirmed	No	Booking already closed

6. State-by-State Reference Table

This combined table shows the rules for both roles side by side at every state.

State	Patient	Caregiver	Notes
pending	Free cancel	Use reject instead	No payment involved
approved	Free cancel	Cancel with reason	Patient not yet paid
confirmed	Cancel + refund	Cancel + penalty flag	Payment already made
on_the_way	BLOCKED	BLOCKED	Point of no return
arrived	BLOCKED	BLOCKED	Caregiver on-site
service_started	BLOCKED	BLOCKED	Service in progress
service_completed	BLOCKED	BLOCKED	Confirm completion only
completed_confirmed	BLOCKED	BLOCKED	Fully closed

7. What Happens When Cancellation Is Blocked

When a cancellation request is rejected by the system, the server returns an HTTP 400 error response. The booking status does not change. Nothing is recorded in the database. The mobile app receives the error message and shows it to the user in an alert dialog.

The key rules the server checks before allowing cancellation:

- Does the booking exist? If not, return 'Booking not found.'
- Is the requester actually a party to this booking? If not, return 'You are not a party to this booking.'
- Is the booking already in a terminal state (cancelled or rejected)?
- Is the current booking status in the hard-blocked list (on_the_way, arrived, service_started, service_completed, completed_confirmed)?
- Is a caregiver trying to cancel a pending booking instead of using reject?
- Is a caregiver submitting a cancellation without a written reason?

All of these checks happen in order. The first check that fails stops the process and returns the relevant error message. If all checks pass, the cancellation is processed.

8. Refund Logic for Card-Paid Bookings

When a patient pays by Mastercard and then cancels a confirmed booking, the system cannot simply close the booking. A refund must be processed through the Mastercard MPGS payment gateway. Here is exactly how that works.

The refund process step by step:

- Step 1 — Patient requests cancellation of a confirmed booking.
- Step 2 — The server checks: was the payment method 'mastercard' and is the booking marked as paid?
- Step 3 — If yes, the booking's refundStatus is set to 'pending' and the cancellation is recorded immediately.
- Step 4 — The refund is triggered against the MPGS gateway using the original transaction reference stored on the booking.
- Step 5 — If the refund succeeds, refundStatus is updated to 'processed'.
- Step 6 — If the refund fails for any reason, refundStatus is set to 'failed' and the case appears in the admin portal for manual processing.

Cash bookings:

If the patient paid by cash, no refund process is needed at all. Cash was never transferred digitally — the patient simply agreed to pay the caregiver on arrival. When a cash booking is cancelled, the status changes to 'cancelled' immediately with no further financial steps required.

Refund status values explained:

Value	What it means
not_applicable	The booking was not paid by card, or no cancellation has been requested. This is the default.
pending	Cancellation was accepted and a refund has been requested. Waiting for the gateway to confirm.
processed	The refund was successfully processed by Mastercard MPGS. The patient's card will be credited.
failed	The refund attempt failed. Admin must process manually through the MPGS merchant dashboard.

9. Penalty Flag for Caregiver Cancellations

When a caregiver cancels a booking that is already in 'confirmed' status, the system raises a penalty flag on that booking record. This is not an automatic suspension or fine — it is a visibility mechanism that allows administrators to monitor caregiver reliability over time.

The penalty flag is stored as `penaltyFlag: true` on the booking document. It is visible in the admin portal under Booking Management. Administrators can filter for flagged cancellations and review a caregiver's full history to decide on appropriate action such as a warning, retraining, or removal from the platform.

What triggers a penalty flag:

- The caregiver cancels a booking that is in 'confirmed' status.
- This means the patient had already paid or confirmed cash payment, and the caregiver is backing out of a firm commitment.

What does NOT trigger a penalty flag:

- Caregiver cancels an 'approved' booking — the patient had not yet paid, so the impact is lower.
- Patient cancels any booking — penalty flags only apply to caregiver cancellations.
- Caregiver rejects a 'pending' booking — rejecting before acceptance is a normal and expected action with no negative weight.

10. What Each Cancellation Records in the Database

Every successful cancellation writes the following fields to the booking document in MongoDB. These are new fields that must be added to the booking schema as part of implementing this feature.

Field name	Data type	What it stores
status	String	Updated to 'cancelled'
cancelledBy	String	Either 'patient' or 'caregiver' — records who initiated the cancellation
cancellationReason	String	The written reason. Required for caregivers. Optional for patients. Null if not provided.
cancelledAt	Date	The exact timestamp when the cancellation was processed by the server
penaltyFlag	Boolean	True only when a caregiver cancels a confirmed booking. False in all other cases.
refundStatus	String	One of: not_applicable, pending, processed, failed. Default is not_applicable.

11. Error Messages the System Returns

All cancellation errors are returned as HTTP 400 responses. The mobile app reads the message field and displays it to the user. Below is every possible error scenario and the exact message the user will see.

Scenario	Message shown to user
Booking does not exist	Booking not found.
User is not patient or caregiver of this booking	You are not a party to this booking.
Already cancelled	This booking has already been cancelled.
Already rejected	This booking was already rejected.
Patient cancels — on_the_way	Cancellation is not allowed — the caregiver is already on the way to you.
Patient cancels — arrived	Cancellation is not allowed — the caregiver has arrived at the location.
Patient cancels — service_started	Cancellation is not allowed — the service is currently in progress.
Patient cancels — service_completed	Cancellation is not allowed — the service has been completed.
Patient cancels — completed_confirmed	This booking is already completed and confirmed.
Caregiver cancels a pending booking	Use the reject action to decline a pending booking request.
Caregiver submits no reason	A cancellation reason is required.
Caregiver cancels — on_the_way or later	You cannot cancel while en route. Please complete the visit.

12. Edge Cases and Special Situations

App goes offline mid-cancellation:

The cancellation is only committed once the server confirms it with an HTTP 200 response. If the network drops after the user taps cancel but before the server responds, the booking status is unchanged. When the app reconnects and resumes polling, it will see the original status and the user can try again safely.

Caregiver has started travelling but has not pressed 'On the Way':

GPS tracking in Nursify starts only when the caregiver explicitly presses the 'On the Way' button, which also changes the booking status. The system does not infer status from GPS data. Therefore if a caregiver has physically started travelling but has not updated the app, the booking remains in 'confirmed' status and cancellation is technically still possible. This is an accepted limitation in the current design.

Both parties try to cancel at the same time:

MongoDB's atomic write operations ensure only one cancellation can succeed. The first request to reach the server updates the status to 'cancelled'. The second request finds the booking already cancelled and receives the error: 'This booking has already been cancelled.'

Refund fails after a patient cancels:

The booking remains cancelled — the cancellation itself is not rolled back if the refund fails. Instead, the refundStatus field is set to 'failed', making the case visible in the admin portal. The administrator can then process the refund manually through the Mastercard MPGS merchant dashboard.

Admin needs to force-cancel a stuck booking:

The current implementation does not include an admin cancellation override in the API. If a booking gets stuck in a live state due to a dispute, an admin must update the booking record directly in MongoDB. A proper admin override endpoint is recommended as a future improvement to the platform.

13. Admin Visibility Over Cancellations

The admin portal's Booking Management section shows all bookings including cancelled ones. Administrators can filter by status to view only cancelled bookings, and can inspect individual booking records to see the full cancellation detail.

What admins can see on a cancelled booking:

- Who cancelled it — whether it was the patient or the caregiver.
- When it was cancelled — the exact date and time.
- The reason provided — the written reason from the caregiver, or from the patient if they chose to supply one.
- Whether a penalty flag was raised — visible when `penaltyFlag` is true on caregiver cancellations of confirmed bookings.
- The refund status — whether a card refund is pending, has been processed, or has failed.

Recommended admin actions:

- Review caregivers with multiple penalty flags and consider issuing warnings or reviewing their approval status.
- Monitor `refundStatus`: failed entries daily and process manual refunds within 24 hours to protect patients.
- Track cancellation rates by caregiver over time to identify unreliable caregivers before they damage patient trust.

Document: Nursify Booking Cancellation Logic | Covers: All booking states, both roles, refund logic, penalty flags, error messages, and edge cases | Intended audience: Developers, product managers, QA testers, administrators